FIRST YEAR PROJECT

GROUP: TIGERS

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https://github.com/LauritsRich/2026-PDS-Tigers.git

PROJECTS IN DATA SCIENCE

1 Introduction

Cutaneous malignancies are some of the most globally spread forms of cancer, with both incidence and mortality rates reaching a continuous increase [1]. Characteristic problems like these occur especially within populations that have a lower value on the Fitzpatrick skin scale. These pathological growths demonstrate a profound variety regarding their histological origins (e.g., melanoma originates from melanocytes, basal cell carcinoma arises from basal cells and squamous cell carcinoma develops from squamous cells).

It is well known that diagnosing pigmented skin lesions is challenging. Dermatologists traditionally rely on the clinical ABCD rule, evaluating features such as Asymmetry, Border, Color and Diameter, to improve diagnostic accuracy [2]. However, visually inspecting a lesion remains fundamentally subjective. To achieve a truly objective diagnosis, this study automates this criterion through computational image processing. Clinical guidelines are translated into feature extraction, measuring border irregularities using geometric metrics and capturing color variability through localized variance.

Our research outlines an application for exploratory skin anomaly evaluation. By computing the chosen ABC features, the software classifies lesions binary as benign or malignant. We conducted our investigation by keeping in mind this central question that guided us: **"How much does the usage of enhanced features improve our model performance?"**. Therefore, our focus was mainly on identifying limitations and working on exceeding them by enhancing our implementations.

2 Related Work

Recent studies provide a clear objective baseline for our thoughts regarding how significant is the correlation between our feature engineering efforts and the results. For example, we found this research utilizing the ISIC2018 dataset, which measured the direct impact of feature enhancement via structural distortion removal. By automating the process where they use an isolation procedure for the structural extraction, they inhibited irrelevant background textures. That helped them to achieve an accuracy of 90.89% [3]. This substantial performance gain directly highlights our hypothesis that the utilization of computationally strengthened features is a major driver of model success.

3 Data

We followed the academic guidelines given, meaning that the PAD-UFES-20 database is the foundation we were given to train our diagnostic architecture. The clinical collection comprises 2,298 clinical captures across 1,373 patients, all characterized by diversified environmental lighting and image dimensions. The dataset categorizes the anomalies contained into six distinct pathologies such as: Actinic Keratosis (ACK), Basal Cell Carcinoma (BCC), Malignant Melanoma (MEL), Melanocytic Nevus (NEV), Squamous Cell Carcinoma (SCC) and Seborrheic Keratosis (SEK).

Because of the extreme lack of MEL lesions (only 52 were found), the algorithm can be biased during training [4]. That's why we simplified the diagnostic categorization into a binary predictive model. Pathology types BCC, SCC and MEL were labeled as malignant, while ACK, NEV and SEK are part of the benign group [5].

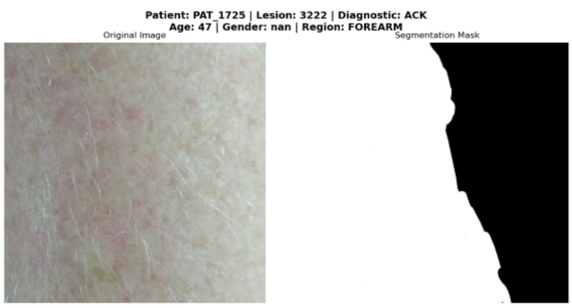
4 Dataset Exploration: PAD-UFES-20

During the data exploration process, we made use of feature extraction to test the quality of the images and the associated masks. Metrics such as asymmetry, border irregularity, and color variation depend on a clearly defined lesion, and we therefore restricted ourselves to images for which there are good segmentation masks. For images where the mask is not satisfactory, feature extraction will produce missing values, and these examples are then dropped from further consideration.

When we applied this process to our 2000 images, a mere 48 images were discarded. It seems that the quality of the data is therefore relatively high. The one disadvantage of this approach is that we assumed that all problems during feature extraction resulted from faulty data, while some images may have been recoverable. Given the very limited amount of data being discarded, however, we are fairly certain that this did not influence the results negatively.

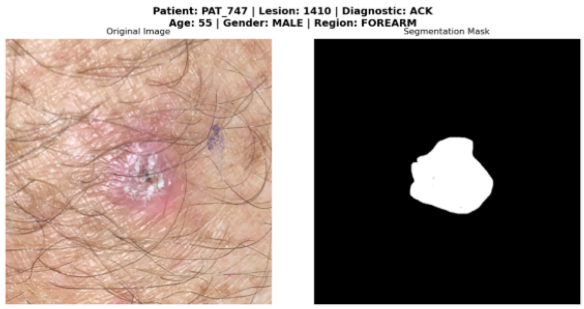
4.1 Annotations & Mask Segmentation

The process of achieving a precise mask segmentation is highly demanding. A study shows that the intensive prelucration required for labeling pixels is a critical need to reduce human annotations [6] and this difficulty is heavily reflected in our dataset. In order to evaluate the precise fit of the provided segmentations, we can face a true challenge, because skin lesions frequently show diffuse, unclear perimeters. As a result, these masks function as approximations rather than absolute ground truths.

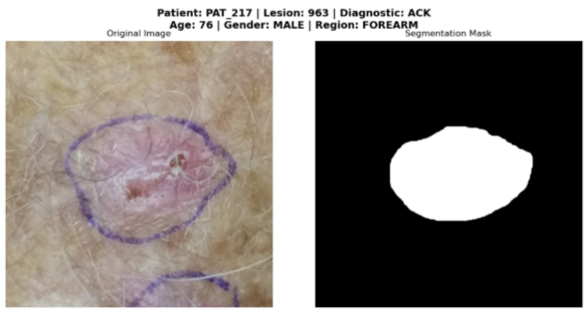


For instance, this image of an Actinic Keratosis lesion caused border ambiguity, diffuse edges and hair obstruction for the mask.

Similar contrast issues persist, as this photo shows a pinkish (close to the skin color) Basal Cell Carcinoma lesion, causing the segmentation process to fail in contouring the entire lesion.



Further complications arise from physical obstructions, because when thick, dark hair covers an ACK lesion, it obscures true borders, forcing the segmentation mask to isolate only the central region while missing the surrounding inflammation.



Beyond biological factors, clinical marks, like the pen ones, further bias the process. These medical markups can distort by causing algorithms to trace artificial ink contour rather than actual biological edges.

Since these images show how such obstructions pervert biological boundaries, human validation is still an essential feature for clinical accuracy in dermatological diagnosis.

5 Baseline Classification Model (ABC)

To establish a diagnostic baseline, Asymmetry, Border and Color features are integrated into one machine-learning model. While individual metrics show statistical differences, their overlap necessitates a combined classifier. This approach provides a reliable, data-driven foundation for automated skin screening by merging structural and chromatic indicators.

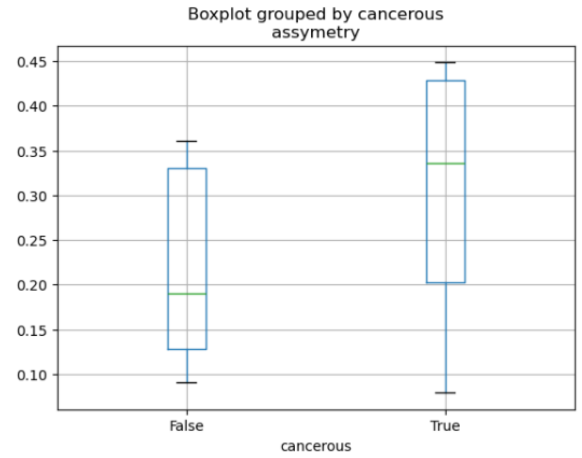
5.1 Feature Extraction & Analysis

5.1.1 Asymmetry

Asymmetry is a primary visual indicator of malignant skin lesions. While benign cases tend to be circular and symmetric, malignant cases frequently develop irregular growth patterns that result in an asymmetric shape [7]. Hence, this visual asymmetry serves as a strong clue that a lesion may be cancerous.

Each mask from the dataset is first loaded and processed using OpenCV. To isolate the region of interest, the mask is cropped to its visible area using NumPy. The asymmetry extraction involves iteratively rotating the mask in 45-degree steps. For each rotation, the mask is flipped and a bitwise XOR operation is applied between the rotated mask and its reflection. The resulting score is then normalized by the original area of the mask. Finally, the mean of these rotational scores is calculated to determine a representative value for lesion asymmetry.

Analysis reveals a distinct difference in asymmetry between benign and malignant skin lesions. Benign cases reveal a median asymmetry score of 0.19 (IQR: 0.13-0.32), whereas malignant cases show a higher median of 0.34 (IQR: 0.20-0.43). While asymmetry is a relevant feature for skin cancer detection, the overlap in scores between benign and malignant cases makes it insufficient as a stand-alone metric. It is important to note that some masks produced scores above 1 (where the XOR-area exceeded the original area). While the underlying cause of this issue is currently unknown, these data points were classified as outliers.



5.1.2 Border

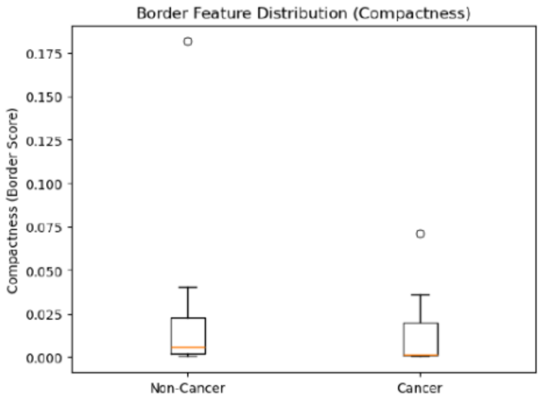
The border feature characterizes the contour regularity of a skin lesion. Clinically, malignant lesions typically present with irregular, asymmetrical and indistinct borders, whereas benign lesions display smooth, circular and well-defined edges. Mathematically, border irregularity differentiates these categories by evaluating metrics such as shape compactness and convexity.

The feature extraction procedure was developed in Python, integrating a suite of specialized libraries [8]. OpenCV (cv2) handled image loading, grayscale conversion, thresholding and hair removal via inpainting and morphological operations (blackhat and tophat). Underlying array manipulations, essential for calculating the total lesion area and summing pixel values, were executed with NumPy. For spatial analysis, binary erosion was applied using the morphology module from scikit-image to approximate the lesion perimeter. Subsequently, convexity analysis was conducted using SciPy's ConvexHull function to determine the lesion's macroscopic perimeter. Finally, to ensure maximum precision in the compactness formula, the constant π from Python's built-in math module was utilized.

Compactness measures a shape's deviation from a perfect circle and is computed as follows:

$$\text{Compactness} = \frac{4\pi A}{L^2}$$

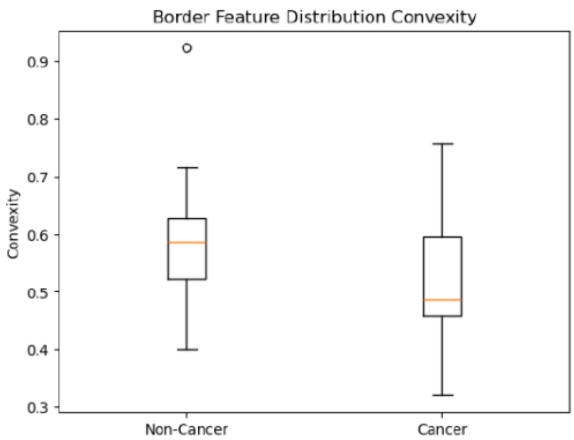
...where A is the area and L is the perimeter. Higher scores indicate a regular, circular structure, while lower scores reflect irregular borders. However, boxplot analysis reveals significant overlap and similar medians between both classes. Consequently, compactness is insufficient as a standalone classifier and must be integrated with other features.



Convexity evaluates contour regularity by comparing a lesion's actual area to the area of its convex hull. It is calculated as:

$$\text{Convexity} = \frac{\text{area of the lesion}}{\text{area of convex hull}}$$

Values approaching 1 denote smooth contours, whereas lower values indicate the irregular, indented edges typical of malignant lesions. Box plot analysis confirms this distinction: non-cancerous lesions consistently exhibit higher convexity (smoother shapes), while cancerous lesions generally produce lower scores due to their structural irregularity.



The extracted border features are highly dependent on the quality of the segmentation mask. Inaccurate segmentation can lead to incorrect area and perimeter estimates, which directly compromise the resulting compactness and convexity values. Additionally, hair removal and thresholding may not reliably isolate the lesion in all cases, especially in images with low contrast or complex backgrounds.

5.1.3 Color

Color variation is one of the most important indicators in the detection of cancerous lesions, as malignant lesions often exhibit multiple irregular colors such as brown, black, red, white, or blue [9]. According to research: "Numerous color descriptors have been applied for the detection, including variation of hues, RGB color channel statistical parameters, color variegation detection using analytical color techniques, spherical color coordinates and (L^*, a^*, b^*) color coordinate features, percentage of the skin lesion containing absolute shades of reddish, bluish, grayish and blackish areas and the number of those color shades present within the skin lesion in dermatoscopy images." [10].

The feature extraction process was implemented in Python using several specific libraries. NumPy was used for array operations, while scikit-image (skimage) was applied for SLIC segmentation and for color space conversions from RGB to HSV and CIE Lab. Additionally, SciPy.stats was used for statistical analysis and Matplotlib was utilized for image array conversion. During extraction, three color spaces were analyzed. The first was the standard RGB color space, which provides useful insights regarding variance in the red channel, a feature that may be linked to malignant lesions. The second was the HSV color space, which is more vigorous under varying lighting conditions. Finally, the CIE Lab color space was included, as research suggests it is clinically relevant because it separates luminance (L^*) from chromatic axes (a^*, b^*), thereby enabling a more perceptually accurate analysis of lesion color distributions [11].

Computations were performed using SLIC superpixel segmentation (10 superpixels). This approach ensures that color features are computed locally for each segment, which aligns with the fact that cancerous lesions exhibit color variegation. This characteristic can then be captured through the variance between values across different segments.

For each of the three color spaces, basic statistical measures (mean and variance) were calculated for each channel. Specifically, for each segment i, these values were computed for the RGB channels (red, green and blue), the CIE Lab channels (L^*, a^*, b^*) and two of the HSV components (saturation and value).

$$\mu_R^i = \frac{1}{N_i} \sum_{p \in S_i} R_p$$

$$\sigma_R^2 = \mathrm{Var}(\{R_p\})$$

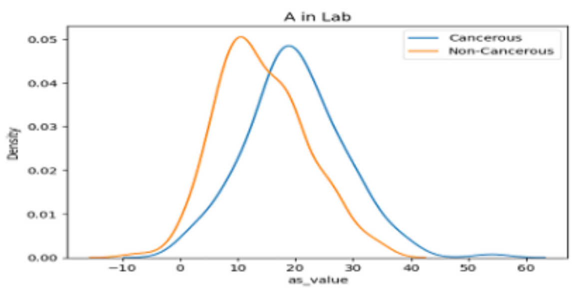
However, hue is circular, so the mean and variance must be circular (although 0.98 and 0.01 appear numerically distant, both correspond to the same color). Hue mean formula:

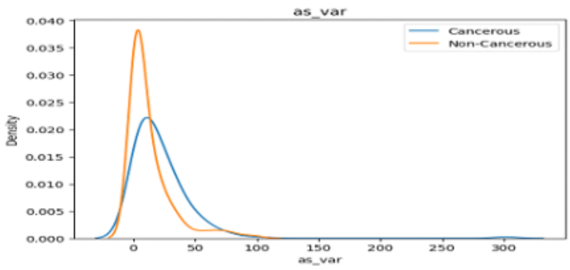
$$\sigma_H^2 = 1 - R, \quad R = \sqrt{(\overline{\cos})^2 + (\overline{\sin})^2}$$

and variance:

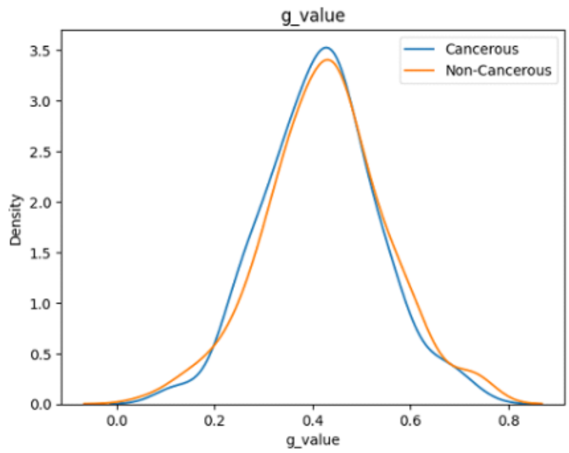
$$\theta = \tan^{-1}\left(\frac{\sum \sin(2\pi H)}{\sum \cos(2\pi H)}\right)$$

To evaluate the collective influence of the color channels, two combined metrics were calculated: mean variance and variance magnitude. KDE plots clearly demonstrate that features such as the CIE Lab a^* channel and its variance (across the green-red color vector) can effectively separate malignant lesions from benign ones.





Other features, such as the RGB green channel, lack predictive value and effectively act as noise during model training.



After evaluating the plots, a promising success could present the following features: as_value*, as_var*, mean_angle_h, s_value, v_value, r_value, bs_var, h_var, s_var, circular_max_min_h.

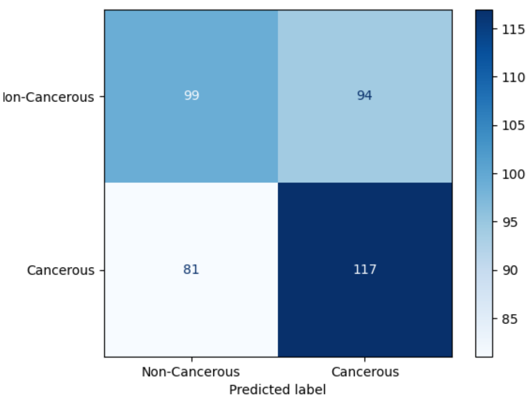
While assessing exhaustive combinations of color features and their statistical variances is beneficial for feature selection, the process is computationally demanding. Consequently, its execution time is substantially longer than that required for morphological features such as asymmetry and border shape.

Moreover, a rich color feature extraction method was implemented to handle exceptions specifically to bypass a ValueError ("more dimensions than allowed") encountered in approximately 3% of the images. The extraction process requires strict spatial alignment between lesion masks and images. Even a 1-pixel mismatch caused failures in segmentation and feature extraction, a profound sensitivity to preprocessing inconsistencies. Additionally, the occurrence of completely empty masks presented further computational risks, potentially triggering errors during SLIC segmentation and variance computation.

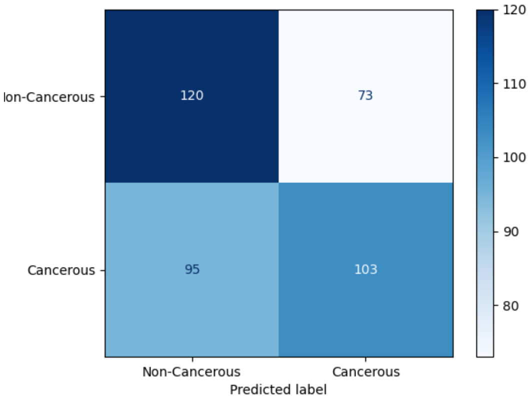
5.2 Best Baseline Model

In order to identify the optimal baseline model, we conducted cross-validation using KNN, Logistic Regression and Random Forest models, varying their respective hyperparameters:

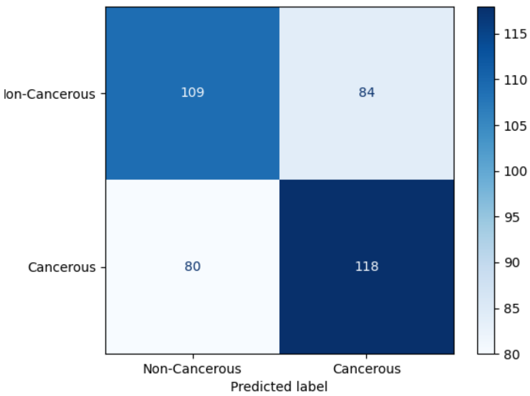
- **KNN**: We adjusted the number of neighbors K, type of weights (uniform and distance) and metric (Euclidean and Manhattan). The optimal parameters chosen were a Euclidean metric, 13 neighbors and uniform weights. This resulted in a best cross-validation AUC score of 0.6443 and a test dataset AUC score of 0.5991. The resulting confusion matrix showed that the model correctly identified 117 cancerous lesions (True Positives) and 99 non-cancerous lesions (True Negatives), while producing 81 false negatives and 94 false positives.



- **Logistic Regression**: Chosen for its fast training capabilities and computational efficiency, this model is well-suited for binary classification baselines [12]. We adjusted the regularization parameter C and the class weights. The optimal parameters utilized an L2 penalty with $C = 10$ (a larger value allowing the model to fit the training data more closely) and balanced class weights (giving importance to minority cancer cases so the model does not ignore them). This led to a best cross-validation AUC score of 0.6652 and a test dataset AUC score of 0.6152. The resulting confusion matrix demonstrated that the model correctly identified 103 cancerous lesions (True Positives) and 120 non-cancerous lesions (True Negatives), while producing 95 false negatives and 73 false positives.



- **Random Forest**: As an ensemble model, it consists of a collection of decision trees trained independently on bootstrapped subsets of the data, with the final prediction made by taking the majority vote [13]. We adjusted the criterion, maximum depth, minimum samples to split and the number of estimators. The optimal parameters chosen were the 'entropy' criterion (which measures data randomness using logarithmic calculations to yield better-balanced trees), a maximum depth of 10 (which cuts the tree in an early stage to prevent it from memorizing noise in the training data), a minimum of 2 samples to split (meaning any node with two samples is forced to split into two individual leaves) and 50 estimators. This resulted in the highest best cross-validation AUC score of 0.6768 and a test dataset AUC score of 0.6456. The resulting confusion matrix showed that the model correctly identified 118 cancerous lesions (True Positives) and 109 non-cancerous lesions (True Negatives), while producing 80 false negatives and 84 false positives.



This was done using a sample size of the data, which included features extracted from 2,000 images. Since our feature extraction process demands segmentation masks, we made sure to include only those images which had corresponding segmentation masks. (Our program checks for the correct pairing.) One restriction with our approach is that we do not create our own masks because we have focused solely on feature extraction.

In building the baseline model, we used basic features without using any sophisticated methods of feature selection. We just chose the most obvious ones: mean and variance of the RGB and HSV color space, asymmetry and border irregularity. One of the purposes of this report is to illustrate that one should take care not only about selecting the proper model but also about properly engineering the features prior to fitting a model. Models such as KNN are very sensitive to noise and poor features can serve as noisy features. The susceptibility to noise makes the models prone to overfitting, which is clearly observable from the scores of our KNN (training score $>$ test score). While Logistic Regression proved more robust than KNN and is great for capturing linear structure in the data, the Random Forest model emerged as the most effective baseline classifier. Furthermore, evaluating a simpler model like Logistic Regression alongside a complex one like Random Forest acts as a safeguard. Significant performance discrepancies can warn us if the complex model has overfitted the training set.

6 Extended Classification Model

The Extended Classification Model integrates the DEFG features: Diameter Irregularity, Evolution Scores, the Fitzpatrick Scale, Fractal Lacunarity and GLCM Grayscale contrast. Because individual features struggle with lack of symmetry, subjective metadata, missing data and dark-skinned patient misclassifications, integrating these metrics creates a vital diagnostic indicator to effectively differentiate cancerous from benign lesions.

6.1 Feature Extraction

6.1.1 Diameter

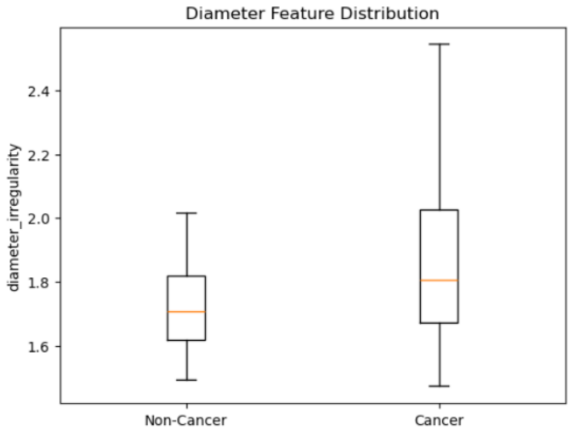
The Diameter feature measures the maximum distance across a skin lesion. Since malignant lesions like melanoma progressively expand while benign ones generally remain small, overall size serves as a vital diagnostic indicator. A diameter exceeding 6mm is recognized as a key clinical warning sign, enabling the model to effectively differentiate cancerous from harmless ones.

The Equivalent Diameter standardizes the lesion's size by calculating the diameter of a circle with an identical area. This feature simplifies the measurement of irregularly shaped growths into a single number. The Equivalent Diameter is calculated using the following formula, where A represents the total area of the lesion in pixels:

$$D_{eq} = \sqrt{\frac{4 \cdot A}{\pi}}$$

Diameter Irregularity is determined by dividing the lesion's maximum diameter by its equivalent diameter. This invariant ratio effectively shows the growth's lack of symmetry and overall shape distortion. Because it is a calculated ratio, it is unaffected by the overall scale of the image. This ensures consistent measurements even if the image is resized [14].

$$\text{diameter irregularity} = \frac{\text{max diameter}}{D_{eq}}$$

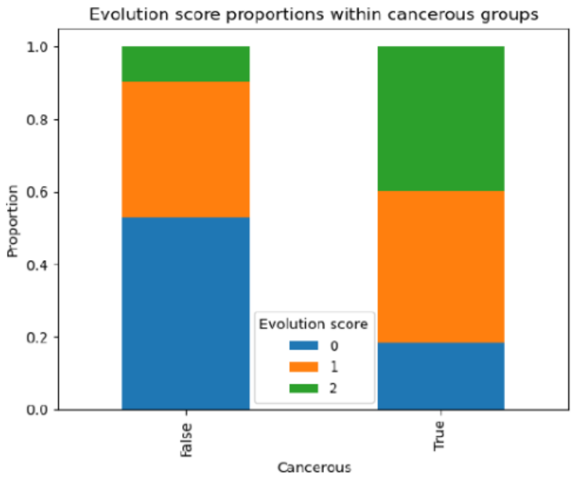


Clinically, increased diameter irregularity correlates with a higher probability of malignancy. Box plot analysis of our data confirms this, showing elevated scores for cancerous lesions. Because there is still an overlap between benign and malignant cases, this metric is most valuable when combined with other features rather than evaluated in isolation. Limitations appear, as the metric captures macroscopic shape but ignores fine border irregularities and internal texture information.

6.1.2 Evolution

Malignant lesions evolve more rapidly than benign ones, making the tracking of these changes over time a critical diagnostic metric [15]. To standardize the diverse affirmative metadata responses (e.g., "yes," "true," "sim"), Pandas was utilized to convert them into binary integers (1 for affirmative, 0 for negative). Summing these values for each case produced a total "Evolution Score".

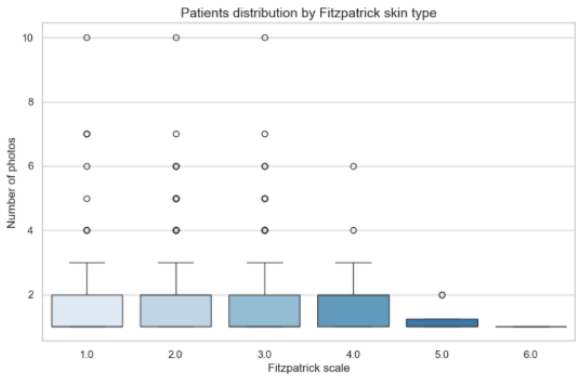
The data reveals that 80% of malignant cases achieve an Evolution Score greater than 0, compared to only 50% of benign cases. Interestingly, this pattern reverses at a score of 2, which occurs in approximately 10% of benign cases but only 0.4% of malignant ones. A key limitation of this feature is that it relies solely on subjective metadata and visual observation rather than automated image analysis. Furthermore, the substantial number of benign cases with elevated scores (> 0) indicates potential overlap, meaning this metric is most effective when integrated with other diagnostic features.



6.1.3 Fitzpatrick

In 1975, the Fitzpatrick Scale was originally designed by the dermatologist with the same name, Dr. Thomas Fitzpatrick, to predict how skin responds to UV exposure, specifically if it is exposed to burning instead of tanning. These days, in aesthetic medicine, it serves as a critical tool to evaluate a patient's UV sensitivity, determine their tolerance for specific procedures and anticipate potential recovery risks [16].

The process begins with data ingestion, where `pd.read_csv` loads the raw metadata into a structured, searchable DataFrame. Next we do the selection phase, where we apply the `.isin()` to isolate only the rows matching the requested patient IDs. We then proceed to do the extraction and formatting, where the `if not filtered.empty:` block ensures the code does not crash if a patient is missing. If a match is found, `.iloc[]` grabs the specific value and an `int()` cast ensures it is returned as an integer number (from 1 to 6).



The boxplot reveals a profound class imbalance shown toward the lower end of the Fitzpatrick scale (1 to 4). The dataset overwhelmingly features patients with a higher predisposition to skin cancer, patient picture distribution across these high-risk categories being relatively uniform. In stark contrast, there is severe data sparsity for darker skin types (5 and 6), which are represented by almost no pictures. An important limiting factor is the missing data within the 'fitzpatrick' column. For a subset of patient records, this information is unrecorded. This lack of data not only reduces the effective sample size for any skin-type-specific analysis but also introduces the risk of exclusion bias.

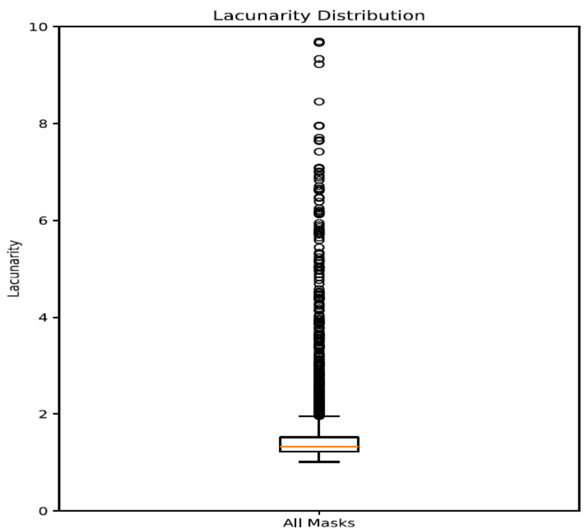
6.1.4 Fractal Lacunarity

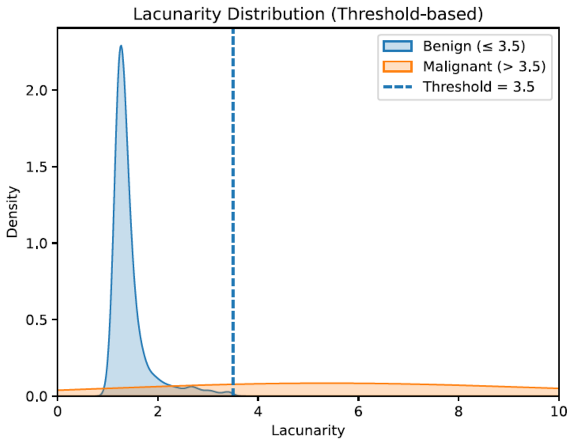
Lacunarity, together with the fractal dimension, is commonly used in the quantitative analysis of dermatological images to characterize the structural and textural properties of malignant skin lesions. Their combination characterizes the complexity of the lesion's geometry and the patchiness in its pigmentation distribution. In image analysis, these images can be processed to compute the quantitative values needed to evaluate the lesion. The resulting metrics offer a numerical estimate of the coarseness of the lesion's color texture, which may assist in differentiating between benign melanocytic nevi and malignant melanoma. In particular, increased lacunarity has been associated with greater irregularity and heterogeneity within the lesion's structure, which is a feature frequently observed in malignant transformation[17]. Together, fractal dimension and lacunarity provide a comprehensive mathematical description of lesion morphology, supporting their use in computer-assisted diagnosis and early melanoma detection.

The lacunarity analysis was implemented on binary skin lesion masks to measure spatial heterogeneity in the lesion's structure. Each grayscale mask was first converted into a binary image using thresholding, where lesion pixels were set to 1 and background pixels to 0. To focus the analysis on the lesion itself and reduce computational cost, each mask was then cropped to its bounding box. Lacunarity was computed using a sliding-window box-counting method accelerated with an integral image, which allows for the fast calculation of pixel sums within each window. A fixed window (box size) was moved across the image, and the number of lesion pixels inside each window was recorded to form a distribution of local densities. Higher lacunarity values indicate greater irregularity and clustering in the lesion structure, while lower values correspond to more uniform distributions.

$$\Lambda = \frac{\sigma^2}{\mu^2} + 1$$

Formula for lacunarity: variance over squared mean plus 1





We can see that in the boxplot most of the lesions are centered at around 1.5, which signifies their benign potential. On the other hand, we can spot a long line of outliers that gets as high as 10, which is a sign of high malignancy. By looking at both, it becomes clear that the "outliers" in the first plot are the primary "malignant" signals in the second; the box plot shows the presence of variation, while the density plot explains its clinical significance.

Despite its usefulness, fractal lacunarity analysis has several limitations when applied to skin lesion masks. One key constraint is its sensitivity to preprocessing steps, such as thresholding and segmentation quality. Any noise, artifacts, or inaccurate lesion boundaries can significantly affect the resulting lacunarity values. Additionally, the choice of box size in the sliding-window method strongly influences the outcome, since different scales may capture different structural characteristics of the lesion, making results less consistent.

6.1.5 Grayscale

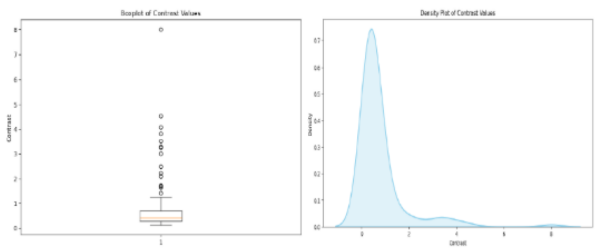
The grey-scale feature shows the relation between the intensity of pixels and is often used in clinical research on skin lesions. According to research published in the Journal of Clinical Medicine, the GLCM contrast parameter serves as a quantitative marker for skin irregularity; specifically, a decrease in contrast (which the study measured at an average of 10.7% following treatment) directly correlates with the reduction of hyperpigmented 'islands' and an increase in the overall homogeneity of the skin surface [18]. Higher contrast values often reflect uneven pigmentation and structural disorder, which are signs of malignant lesions such as melanoma, while lower, uniform values typically represent benign lesions.

The gray-scale extraction function was implemented in Python using `os` (for folder accessing), NumPy (for array manipulation and computing statistical measures) and Scikit-image (for image conversion and building the GLCM). The contrast feature in this project is computed using the Gray Level Cooccurrence Matrix, which measures how often pairs of pixel intensities occur at specific spatial relationships.

To simplify analysis, images are first converted to grayscale, reducing color information to a single intensity value per pixel. This focuses exclusively on variations in brightness rather than color. Next, the grayscale image is simplified by reducing the number of gray levels, from 256 to 32, to create a more manageable matrix. The GLCM is then constructed by tracking how often a pixel of intensity 'i' occurs at a specific distance and angle from a pixel of intensity 'j'. Common angles include 0°, 45°, 90°, and 135° and the resulting matrix is normalized to treat the counts as probabilities.

Finally, the contrast feature is calculated from the GLCM, giving higher weight to pixel pairs with large intensity differences. High contrast values (3.8-31.0) indicate irregular textures, often associated with cancerous lesions, while low values (0.05-2.0) suggest smoother textures typical of benign regions. Values in between represent borderline cases.

$$\text{Contrast} = \sum_{i,j=0}^{N_g-1} P(i,j)(i-j)^2$$

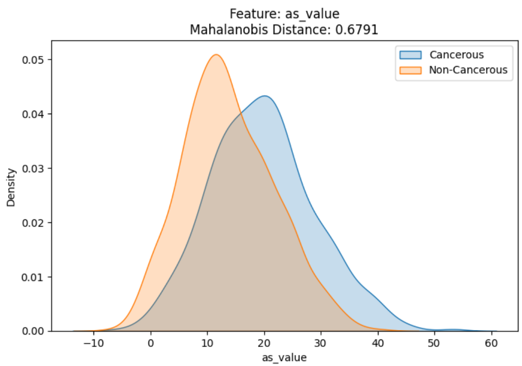


The code runs efficiently as it filters all the images given to focus primarily on the light-skinned ones. When processing images of dark-skinned patients, the algorithm, depending on the brightness of the lesion, fails to differentiate the grayscale variation of the lesion from the surrounding skin. This leads to an erroneously low contrast value, which results in a false conclusion of whether this is a benign or malignant lesion.

6.2 Best Extended Model

Features in the model were chosen after intensive explorations and univariate analysis. The goal was to assess various parameters in order to detect the most promising predictors. While the relevance of each feature was hard to judge right away, we took an initial set of 30 features and then narrowed them down to 13, excluding clearly irrelevant features.

To achieve our objective, we extracted 30 features from the data set so that we could choose the best ones manually. Although this method was quite expensive for computation, studies guided us towards some unique features that would help us improve the performance of our extended model. For determining the best features, we started by measuring the distance between the distributions using the Mahalanobis distance on our KDE graphs. Next, we compared this information with the correlation matrix to rule out redundant features. After completing this step, we ended up with only 13 features, which improved our overall model performance.

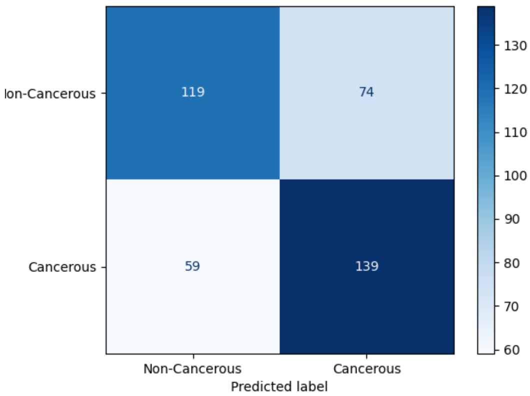


(See section 12 Appendix, subsection 6.2.1 for the image related.)

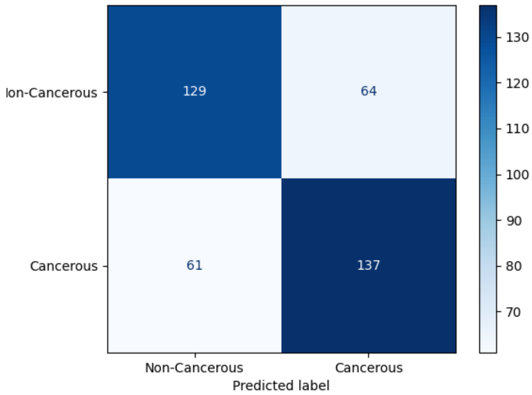
As for KDE plots, while useful for detection of single effect, this method may miss the interdependencies between different features. For this reason, we used a combination of different measures for color features, though mostly all other features were taken separately.

Using the extended feature set, we re-evaluated our models to observe performance gains across three classifiers:

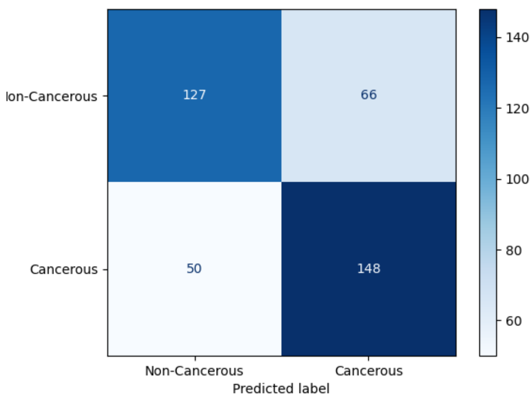
- **KNN**: The optimal hyperparameters selected were a Euclidean metric, 13 neighbors and uniform weights. While KNN is generally sensitive to high-dimensional noise and prone to overfitting, the extended model achieved an improved cross-validation AUC of 0.7598 and a test dataset AUC of 0.7139. The resulting confusion matrix revealed that the model correctly identified 139 cancerous lesions (True Positives) and 119 non-cancerous lesions (True Negatives), while producing 74 false positives and 59 false negatives.



- **Logistic Regression**: We adjusted the regularization parameter C and the class weights. The optimal parameters chosen were an L2 penalty with $C = 0.1$ (applying stronger regularization to keep weights small and prevent overfitting on the expanded feature set) and balanced class weights. This model achieved a superior best cross-validation AUC score of 0.7862 and a test dataset AUC score of 0.7429. The resulting confusion matrix demonstrated that the model correctly identified 137 cancerous lesions (True Positives) and 129 non-cancerous lesions (True Negatives), while producing 64 false positives and 61 false negatives.



- **Random Forest**: We adjusted the criterion, maximum depth, minimum samples to split and the number of estimators. The optimal parameters chosen were the 'entropy' criterion, a maximum depth of 10, a minimum of 5 samples to split (meaning a node with 4 samples cannot split and must become a leaf, which prevents the creation of highly specific rules and curtails overfitting) [19], and an increased 200 estimators (which further reduces the risk of overfitting at the cost of increased running time and memory usage). This model achieved a best cross-validation AUC score of 0.7848 and the highest test dataset AUC score of 0.7699. The resulting confusion matrix demonstrated that the model correctly identified 148 cancerous lesions (True Positives) and 127 non-cancerous lesions (True Negatives), while producing 66 false positives and 50 false negatives.



While the extended features improved overall AUC scores across the board, the Random Forest model emerged as the most effective classifier. This gap in performance highlights a fundamental limitation of Logistic Regression: it assumes a linear decision boundary and struggles when relationships between texture, shape and cancer probability are non-linear or contain interaction effects [20]. Additionally, Logistic Regression can exhibit weaker probability calibration near the extremes (0 and 1). In a clinical environment, a falsely confident near-zero probability prediction could cause a malignant lesion to be disastrously dismissed. However, even the 50 false negatives produced by the Random Forest represent a highly risky mistake, as the presence of a malignancy goes unnoticed. In order to address such issues in the course of diagnosis and treatment, rather than limiting oneself to regular feature selection, the use of class weightings—such as the 'balanced' weights utilized in our Logistic Regression model, can be applied to heavily penalize the algorithm for missing malignant cases.

6.3 Example of how we chose our hyperparameters based on (Random Forest)

To finalize the hyperparameters for our extended Random Forest model, we utilized a grid search approach. However, instead of allowing the algorithm to blindly default to the combination with the absolute highest score, we outputted the top 10 parameter combinations and conducted a manual selection. When evaluating these top combinations, we analyzed both the mean cross-validation AUC scores and their corresponding standard deviations. Our selection logic was driven by the need for a generalizable model, meaning we actively tried to avoid choosing hyperparameters from the "edge" of our grid that might encourage overfitting. For instance, while the top-ranked combination achieved a slightly higher mean AUC of 0.7851, it relied on the absolute minimum sample split of 2 and exhibited a higher standard deviation of 0.0202. We intentionally opted for the second-best combination, which utilized the 'entropy' criterion, 200 estimators, a maximum depth of 10 and a safer minimum of 5 samples to split. Although its mean AUC was marginally lower at 0.7848, it demonstrated much greater stability across data folds with a lower standard deviation of 0.0174. By manually stepping back from the extreme edges of the parameter grid, we ensured our final model was optimized for consistency rather than just a peak training score.

(See section 12 Appendix, subsection 6.3.1 for the image related.)

7 Evaluation and Results

The central question guiding this research was: **"How much does the usage of enhanced features improve our model performance?"**. Based on the empirical data gathered across all three classifiers, the integration of enhanced features produced substantial and measurable improvements in diagnostic accuracy.

In the baseline evaluation, utilizing only ABC features, the models struggled to establish a distinct decision boundary between benign and malignant lesions. AUC scores stagnated between 0.5991 (KNN) and 0.6456 (Random Forest). These lower scores reflected the inherent ambiguity of relying solely on basic structural and chromatic indicators, which often overlap between cancerous and non-cancerous skin anomalies.

Following the introduction of our enhanced feature set, every model demonstrated a drastic performance leap:

- **KNN** saw its test AUC jump from 0.5991 to 0.7139. While it remained our weakest model and continued to show signs of overfitting due to its sensitivity to high-dimensional noise, the enhanced features provided enough spatial separation to improve its baseline accuracy significantly.

- **Logistic Regression**: the AUC test improved from 0.6152 to 0.7429. The use of an L2 penalty and balanced class weights allowed it to effectively interpret the new features. However, its assumption of a linear decision boundary ultimately restricted its ability to capture the complex, non-linear interactions between texture, shape and metadata.

- **Random Forest** emerged as our champion model, with its test AUC rising from 0.6456 to 0.7699. As a whole method, it proved highly capable of processing the diverse data types in our extended set without memorizing noise.

Beyond standard accuracy metrics, a dermatological screening tool must be evaluated on its clinical safety—specifically, its False Negative rate. A false negative is a critical failure, as it results in a malignant tumor being misclassified as benign, delaying potentially life-saving treatment.

Our feature engineering efforts directly mitigated this risk. In the baseline Random Forest model, the algorithm produced 80 false negatives. By narrowing our initial 30 extracted features down to the 11 most optimal predictors (utilizing Mahalanobis distance on KDE graphs and correlation matrices), the extended Random Forest model successfully reduced the number of false negatives to 50, while simultaneously increasing True Positives from 118 to 148.

The results definitively prove our hypothesis: computationally strengthened features are a major driver of model success. The transition from subjective visual approximations to quantitative structural, textural and historical data allowed the Random Forest classifier to achieve a highly respectable test AUC of 0.7699. However, while the reduction in false negatives is promising, 50 missed malignancies out of our test set indicate that the model is not yet ready for autonomous clinical deployment, being just a prototype. It currently serves best as a highly effective exploratory or assistive screening tool that flags high-risk lesions for mandatory human validation.

8 Open Question: "Does Model Ensembling improve performance?"

A central open question in our research was whether combining the predictions of our three individual models (KNN, Logistic Regression and Random Forest) could give a better overall performance than any single model alone. The underlying hypothesis was that an ensemble pipeline could counter the individual flaws of each classifier by combining their strong points. For instance, if KNN struggled with specific lesion records, the Random Forest or Logistic Regression models might be able to step in and correct the final decision. To test this theory, we implemented and evaluated two distinct ensembling methods: Maximum Certainty and Power-Weighted Average.

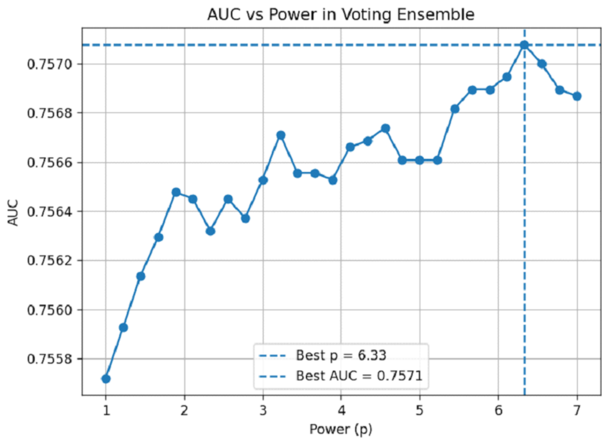
The Maximum Certainty method values absolute confidence over general harmony. We operated under the assumption that consensus can sometimes dilute the truth and in certain clinical cases, we want a highly confident "specialist" to override an uncertain crowd. Mathematically, we evaluated the distance of each model's prediction probability (P) from total randomness (0.5) using the following formula:

$$D = |P - 0.5|$$

The absolute value is taken because the distance cannot be negative. After calculating this distance for each model, the values are compared. The model with the furthest distance from 0.5 wins, and its prediction probability represents the entire group. For example, if the models produce probabilities resulting in distances of 0.14 for KNN, 0.28 for Logistic Regression and 0.38 for Random Forest, the algorithm selects the Random Forest's prediction because it demonstrates the highest degree of certainty. By relying on this method, we prevent the final prediction probability from dropping due to simple averaging.

Alternatively, because standard voting systems often fail to weigh model confidence correctly, a major risk when dealing with critical diagnoses like skin lesions, we also tested a Power-Weighted Average (exponential blend). The primary purpose of this mathematical method is to silence uncertainty and amplify confidence through nonlinear scaling. Instead of treating all model percentages equally, the algorithm alters the probabilities to strictly punish hesitation and reward certainty.

The final combined probability (P_{final}) is derived using a fractional formula where the numerator calculates the total nonlinear scaling favoring the malignant classification (Class 1). The denominator sums up the total scaling for both the malignant and benign (Class 0) outcomes. This standardization ensures the final combined score scales gracefully into a valid probability between 0 and 1. To evaluate this systematically, we ensured all three models predicted on the exact same images using a fixed random state, calculated P_{final}, and applied a standard 0.5 threshold to classify the lesion against the actual diagnosis.



Despite its mathematical robustness, the Power-Weighted Average introduced notable clinical limitations:

- **The Overconfidence Trap:** The logic is built entirely on the assumption that high confidence equals high accuracy. If one model is highly confident but incorrect, the formula blindly trusts it and forces a misclassification, completely ignoring the valid disagreement of the other two models.

- **Prediction Clashes:** When two models are highly confident about opposite outcomes, the formula mathematically clashes. The opposing weights cancel each other out, leaving the final probability stuck near the 0.5 middle ground, effectively reducing the diagnosis to pure guessing.

The fundamental objective of our open question was to observe whether the combination of our three models is essentially better than the prediction of each one separately. Ultimately, neither the Maximum Certainty nor the Power-Weighted Average method significantly boosted our AUC.

The lack of improvement indicates that our classifiers are fundamentally struggling with the exact same difficult records. Because the models share the same blind spots, combining them does not generate any new predictive power.

9 Conclusion

At the start of this project, we proceeded by handling a challenging medical problem: objectively detecting and classifying skin cancer. As we worked through the data, it became clear that moving away from subjective visual guesses and relying on computational feature extraction makes a massive difference. When we built our baseline model using just the standard ABC metrics, it really struggled to draw a clear line between benign and malignant lesions. But once we brought in enhanced features, like Diameter Irregularity, Fractal Lacunarity and GLCM Grayscale contrast, things turned around. We ended up not using the Evolution and Fitzpatrick as part of our features, because they are using data from the metadata file, which is not a reliable source in analyzing our problem, but we still included the process we used to decide that. Our Random Forest model hit a much stronger test AUC of 0.7699.

Even more importantly, this deep dive into feature engineering helped us cut our false negatives down from 80 to 50. We also experimented with combination modeling, hoping that merging different classifiers might cover up their individual weak spots. As it turns out, we proved to ourselves that good feature engineering beats model ensembling hands down. Grouping the models together didn't give us a magical performance boost, simply because they all shared the exact same analytical blind spots.

While we are proud of the improvements we made, we have to be realistic about the flaws. Missing 50 malignant cases out of 391 in a test set is still way too high of a risk for the real world. Because of this, we know our current model isn't ready to be an autonomous, reliable diagnostic source. However, it does a great job as a prototype assisting tool, one that can flag high risk lesions and point dermatologists exactly where they need to look.

9.1 Future improvements

Looking back at everything we have built, our biggest lesson is that we should not spend future efforts trying to "fix" bad predictions with overly complex model building or fancy ensembling tricks. If we want better accuracy, we need to spend our energy on advanced feature engineering. In the end, feeding the model better data is far more effective than trying to manipulate the outputs.

If we were to take this project to the next level, keeping things grounded, here is where we would focus:

- **Cleaning up the segmentations:** Faulty segmentations are our biggest obstruction. We need to find better ways to handle borders, lesions that blend into the skin and physical obstacles like thick hair or clinical pen marks. Getting cleaner outlines will directly improve all of our anatomic metrics.

- **Diversifying the dataset:** One obtrusive issue we found during data exploration was a severe lack of diversity, especially regarding darker skin tones (Fitzpatrick types 5 and 6). To make this tool truly useful and fair for everyone, we desperately need a more globally representative dataset so the model does not just work well for patients with high UV sensitivity.

- **Exploring Deep Learning:** For this project, we intentionally limited our scope to the foundational machine learning techniques and feature extraction methods taught in this course. While this provided an excellent, structured baseline, future work could expand beyond the current curriculum to implement deep learning. Transitioning to models like Convolutional Neural Networks (CNNs) would allow the system to autonomously learn abstract visual patterns directly from the raw pixels, moving past the constraints of what we can manually code.

10 List of Abbreviations

ABC / ABCD Asymmetry, Border, Color, (Diameter) criteria

ACK Actinic Keratosis

AUC Area Under the ROC Curve

BCC Basal Cell Carcinoma

CIE Lab Commission Internationale de l'Éclairage (L^*, a^*, b^*) color space

CNNs Convolutional Neural Networks

CSV Comma-Separated Values

DEFG Diameter, Evolution, Fitzpatrick, Fractal Lacunarity, GLCM features

GLCM Gray-Level Co-occurrence Matrix

HSV Hue, Saturation, Value color space

IQR Interquartile Range

ISIC International Skin Imaging Collaboration

KDE Kernel Density Estimation

KNN K-Nearest Neighbors

MEL Malignant Melanoma

NEV Melanocytic Nevus

PAD-UFES-20 Pathology Dermatology - Universidade Federal do Espírito Santo dataset

RGB Red, Green, Blue color space

SCC Squamous Cell Carcinoma

SEK Seborrheic Keratosis

SLIC Simple Linear Iterative Clustering

UV Ultraviolet radiation

XOR Exclusive OR (bitwise logic gate)

11 Bibliography

[1] A. H. Roky, M. M. Islam, A. M. F. Ahasan, M. S. Mostaq, M. Z. Mahmud, M. N. Amin and M. A. Mahmud: ”Overview of skin cancer types and prevalence rates across continents” from Cancer Pathogenesis and Therapy, vol. 3, no. 2, pp. 89-100, Aug. 2024.

[2] F. Nachbar, W. Stolz, T. Merkle, A. B. Cognetta, T. Vogt and M. Landthaler: ”The ABCD rule of dermatoscopy: High prospective value in the diagnosis of doubtful melanocytic skin lesions” from Journal of the American Academy of Dermatology, vol. 30, no. 4, pp. 551-559, Apr. 1994.

[3] K. Gencer: ”Enhanced Skin Lesion Classification through Hair Artifact Removal and Transfer Learning Models” from Black Sea Journal of Engineering and Science, vol. 8, no. 4, pp. 1007-1021, Jul. 2025.

[4] S. Das: ”Automated Bias Reduction in Deep Learning Based Melanoma Diagnosis using a Semi-Supervised Algorithm” from medRxiv, 2021.

[5] T.-L. Lin, K.-H. Lee, R. Karmakar, A. Mukundan, J. Sundarraj, C.-T. Lu, S.-C. Hsieh and H.-C. Wang: ”Artificial Intelligence-Assisted Dermatologic Screening: Epidemiology and Clinical Features of Basal Cell Carcinoma, Squamous Cell Carcinoma, Seborrheic Keratosis and Actinic Keratosis” from Bioengineering, vol. 12, no. 11, p. 1258, 2025.

[6] T. Siriborvornratanakul: ”Reducing Human Annotation Effort Using Self-supervised Learning for Image Segmentation” from Artificial Intelligence in HCI, Lecture Notes in Computer Science, vol 14734. Springer, Cham, 2024.

[7] Pacheco, 2020. PAD-UFES-20: ”A skin lesion dataset composed of patient data and clinical images collected from smartphones.” Data in Brief, 32, p. 106221.

[8] S. S. Al-Rawi: "A comprehensive study on skin cancer detection using image processing" PeerJ Computer Science, vol. 6, p. e268, 2020.

[9] M. E. Celebi, H. Iyatomi, G. Schaefer and W. V. Stoecker: "Computerized Analysis of Pigmented Skin Lesions: A Review" from International Journal of Biomedical Imaging, vol. 2011, Art. no. 582426, 2011.

[10] M. E. Celebi, Y. A. Aslandogan, R. J. Stoecker, H. Iyatomi, H. Oka and M. Tanaka: "Unsupervised border detection of dermoscopy images" from Skin Research and Technology, vol. 13, no. 4, pp. 454-462, 2007.

[11] M. M. M. Al-Hatab, A. S. I. Al-Obaidi and M. A. Al-Hashim: "Exploring CIE Lab Color Characteristics for Skin Lesion" from Technical Engineering College, Northern Technical University and Department of Computer Science, College of Computer Science and Mathematics, University of Mosul, Iraq, 2024.

[12] GeeksforGeeks: "Understanding Logistic Regression"

[13] Atlassian: "Decision tree" from Atlassian Work Management.

[14] E. Nasr-Esfahani: "Classification of Skin Cancer Using Artificial Neural Network" from Computational and Mathematical Methods in Medicine, vol. 2016, Article ID 8934242, 2016.

[15] Cancer Research UK: "Symptoms of melanoma skin cancer" from Cancer Research UK, n.d.

[16] Medicreations: "The Fitzpatrick Scale: What Every Esthetician Should Know" from Medicreations Blog, Jun. 16, 2025.

[17] S. S. Cross, A. J. G. McDonagh, T. J. Stephenson, D. W. K. Cotton and J. C. E. Underwood: "Fractal and Integer-Dimensional Geometric Analysis of Pigmented Skin Lesions" from The American Journal of Dermatopathology, 1995.

[18] I. Wawrzyk-Bochenek, M. Rahnama, S. Wilczyński and A. Wawrzyk: "Quantitative Assessment of Hyperpigmentation Changes in Human Skin after Microneedle Mesotherapy Using the Gray-Level Co-Occurrence Matrix GLCM) Method" from Journal of Clinical Medicine, vol. 12, no. 16, p. 5249, 2023.

[19] A. Jain: "Everything about Random Forest" from Medium.

[20] GeeksforGeeks: "Advantages and Disadvantages of Logistic Regression"

12 Appendix

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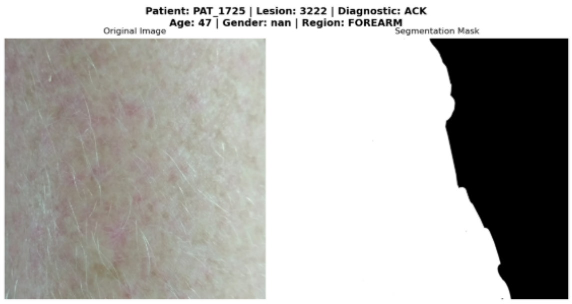

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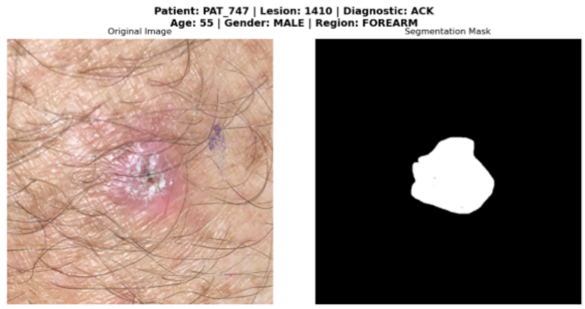
12.3 Figure 4.3



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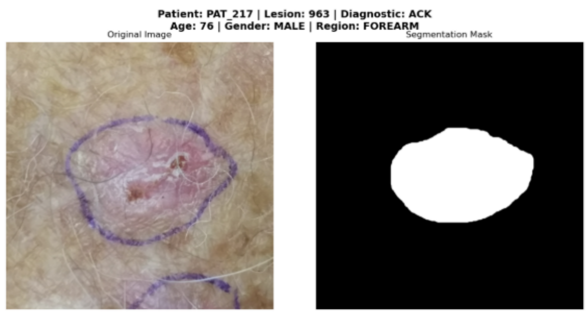

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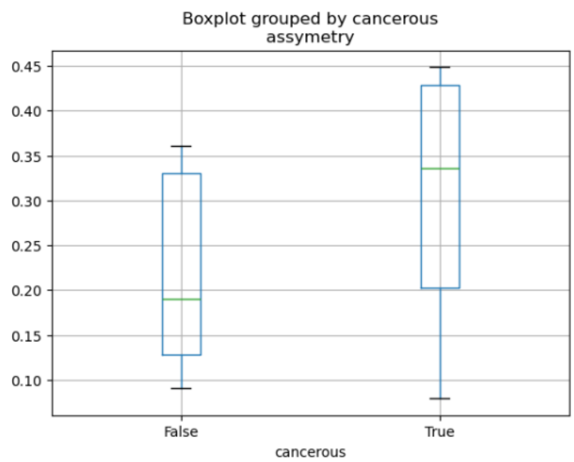

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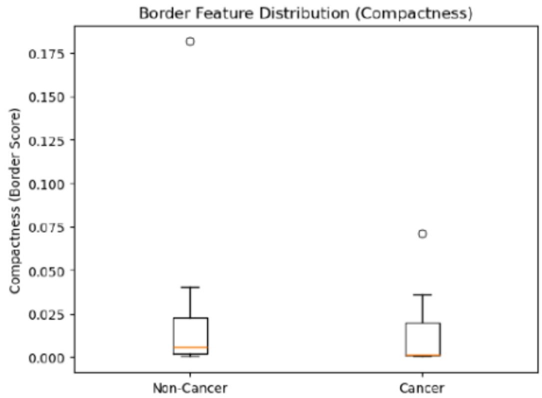

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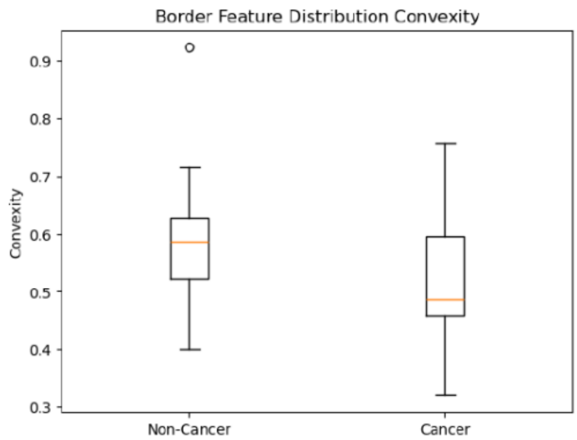

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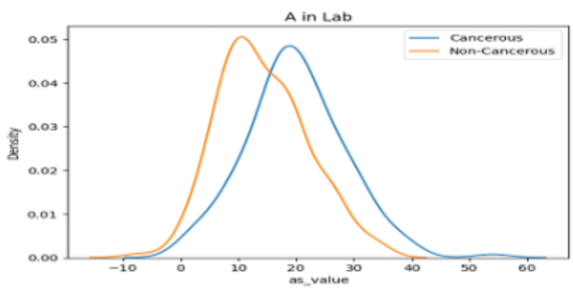

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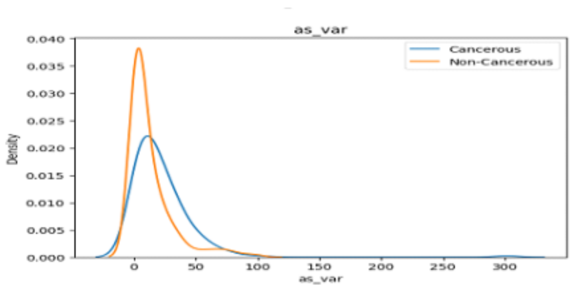

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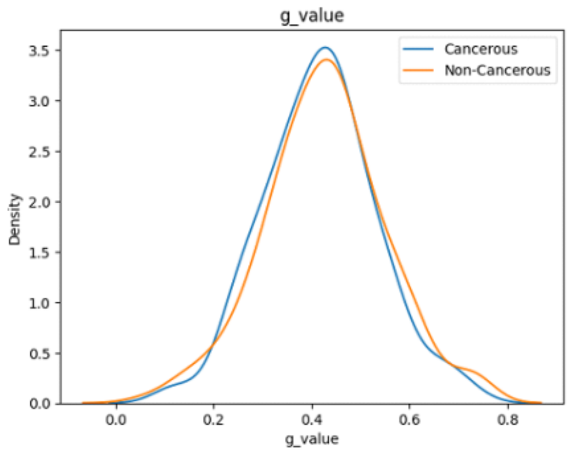

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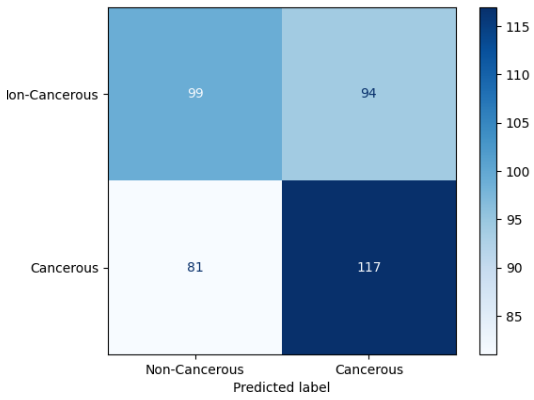

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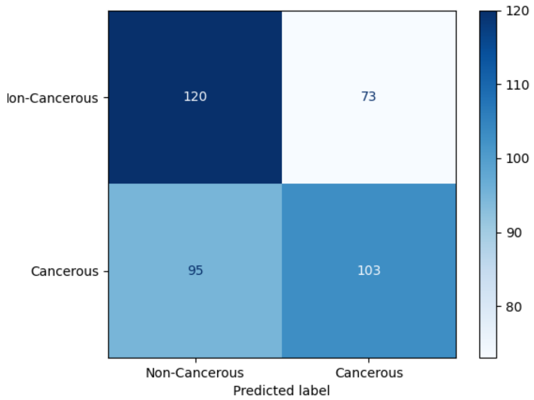

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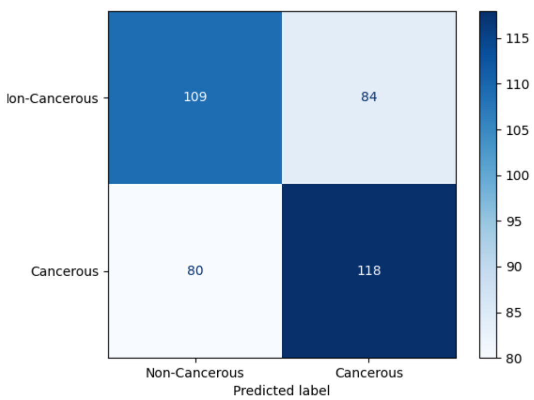

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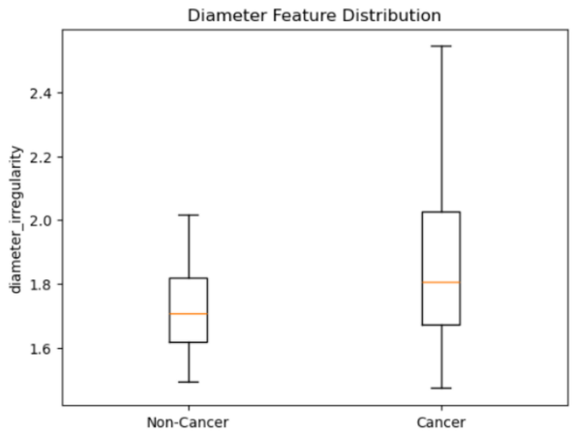

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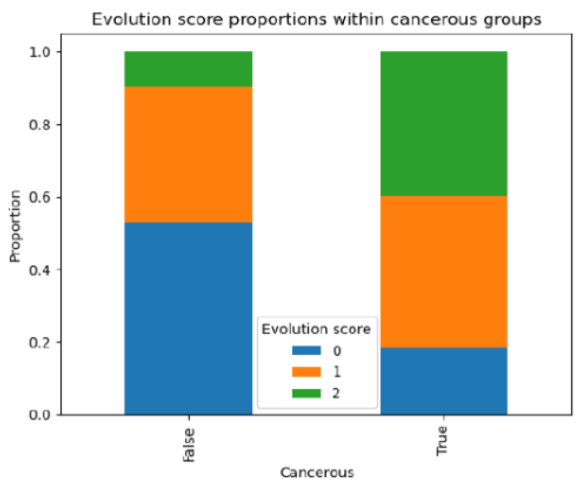

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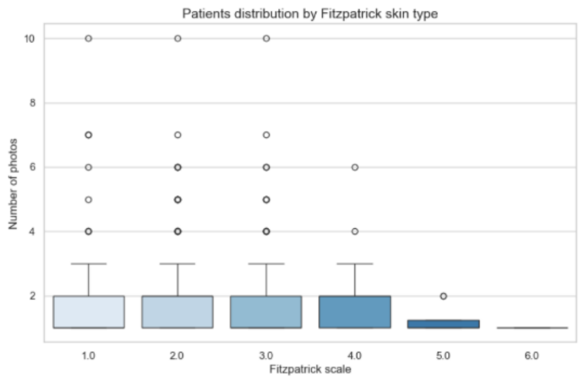

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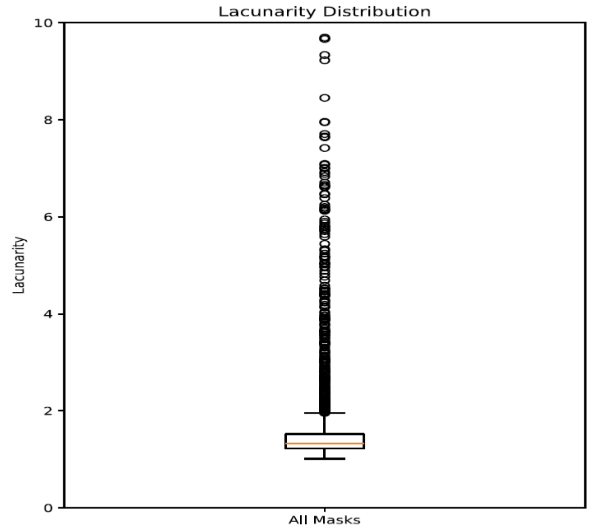

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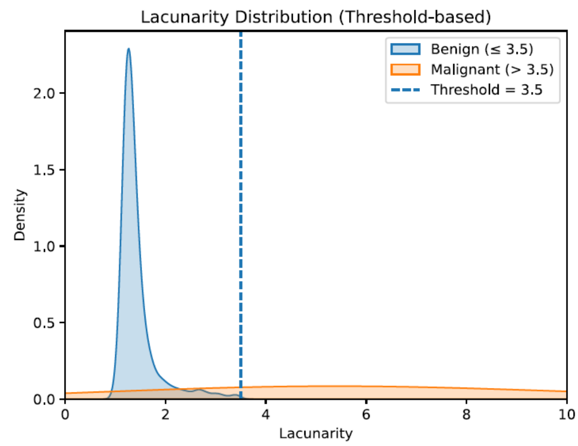

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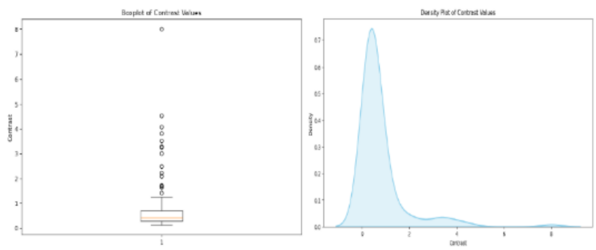

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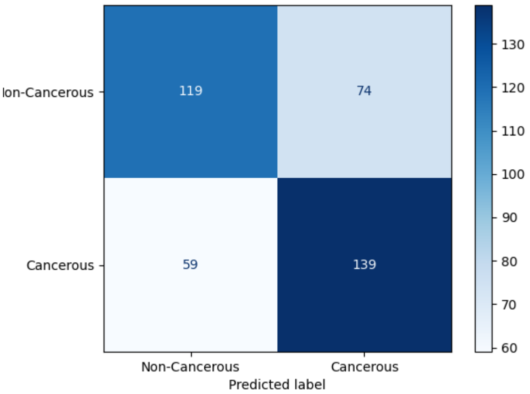

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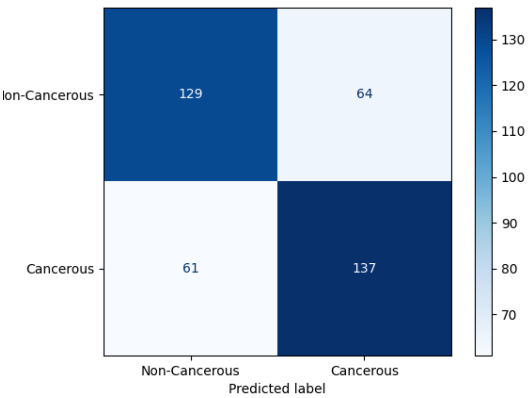

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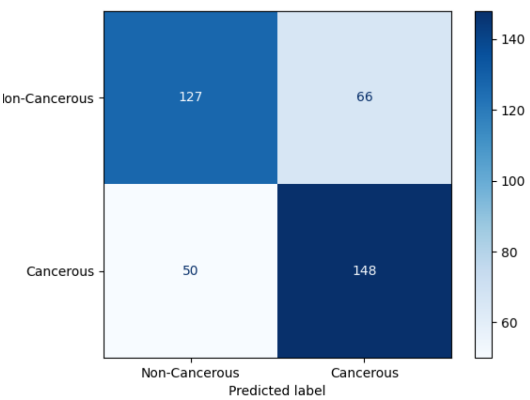

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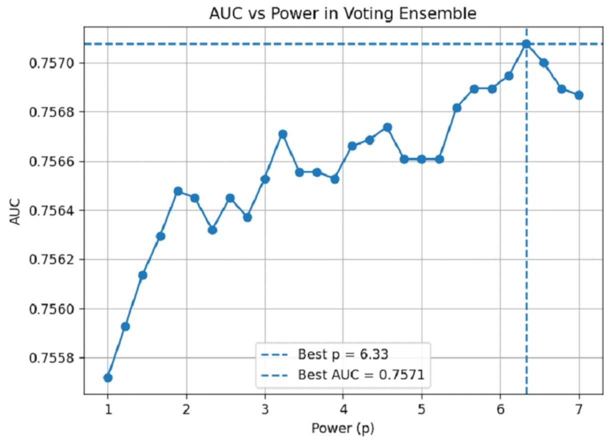

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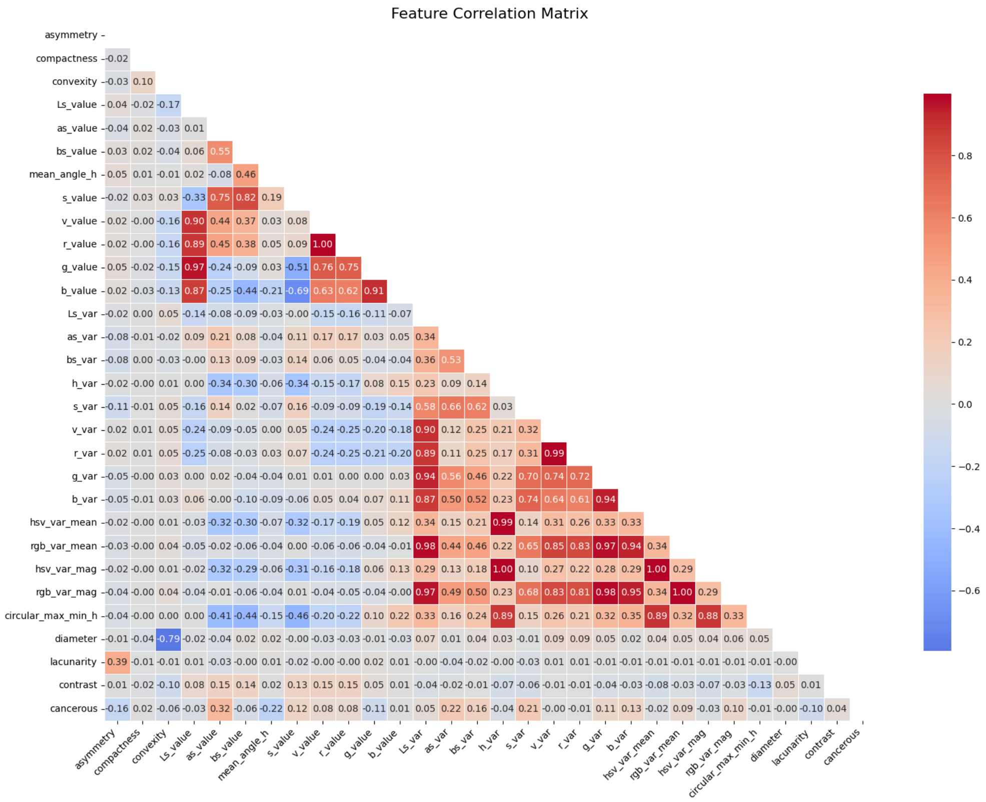

See section 6.2 for context.

12.25 Figure 6.3.1

param_rf__n_estimators	param_rf__max_depth	param_rf__min_samples_split	param_rf__criterion	mean_test_score	std_test_score
100	10	2	gini	0.7851	0.0202
200	10	5	entropy	0.7848	0.0174
200	10	2	gini	0.7846	0.0184
50	10	5	entropy	0.7843	0.0120
200	None	5	entropy	0.7842	0.0177
100	10	5	gini	0.7841	0.0156
200	10	2	entropy	0.7840	0.0188
200	10	5	gini	0.7839	0.0165
100	10	5	entropy	0.7839	0.0190
50	20	5	gini	0.7833	0.0126

See section 6.3 for context.